

## Homework Set HW5 due 2/23/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt) Problem 2, Section 3.3 page 125.

If  $f(x) = 2 \sin x + \tan x$ , then  $f'(x) =$  \_\_\_\_\_

2.(1 pt) Problem 4, Section 3.3 page 125.

If  $f(x) = 2 \sec(x) + 3 \tan(x) - \tan(\pi/3)$ , then  $f'(x) =$  \_\_\_\_\_

3.(1 pt) Problem 6, Section 3.3 page 125.

If  $f(x) = \cos^2(x)$ , then  $f'(x) =$  \_\_\_\_\_

4.(1 pt) Problem 8, Section 3.3 page 125.

If  $f(x) = 2x^3 \sin(x) - 3x \cos(x)$ , then  $f'(x) =$  \_\_\_\_\_

5.(1 pt) Problem 10, Section 3.3 page 125.

If  $f(x) = (x^2 + 2) \sin(x)$ , then  $f'(x) =$  \_\_\_\_\_

6.(1 pt) Problem 12, Section 3.3 page 125.

If  $f(x) = \frac{e^x}{\sin(x)}$ , then  $f'(x) =$  \_\_\_\_\_

7.(1 pt) Problem 14, Section 3.3 page 125.

If  $f(x) = \frac{\sec(x)}{2 - \cos(x)}$ , find  $f'(x) =$  \_\_\_\_\_

8.(1 pt) Problem 16, Section 3.3 page 125.

If  $f(x) = \frac{\ln(x)}{x^2}$ ,  
 $f'(x) =$  \_\_\_\_\_

9.(1 pt) Problem 18, Section 3.3 page 125.

If  $f(x) = \frac{\ln(x)}{x}$ , then  $f'(x) =$  \_\_\_\_\_

10.(1 pt) Problem 20, Section 3.3 page 125.

If  $f(x) = \frac{x \cos(x)}{e^x}$ , then  $f'(x) =$  \_\_\_\_\_

11.(1 pt) Problem 26, Section 3.3 page 125.

If  $f(x) = \frac{x}{1 - \sin(x)}$ , find  $f'(x) =$  \_\_\_\_\_

12.(1 pt) Problem 44, Section 3.4 page 136.

Let  $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$  denote the position of an object subject to a constant downward acceleration due to gravity  $g$  (where  $g$  is given approximately by  $32 \text{ ft/sec}^2$  or  $9.8 \text{ m/sec}^2$ ,  $v_0$  is the initial velocity and  $s_0$  is the initial displacement.

For this problem, a ball is thrown vertically upward from the ground with an initial velocity of 160 ft/sec.

(a) When will the ball hit the ground? \_\_\_\_\_ sec.

(b) With what velocity will the ball hit the ground?  
\_\_\_\_\_ ft/sec.

(c) When will the ball reach its maximum height?  
\_\_\_\_\_ sec.

13.(1 pt) Problem 41, Section 3.4 page 136.

Let  $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$  denote the position of an object subject to a constant downward acceleration due to gravity  $g$  (where  $g$  is given approximately by  $32 \text{ ft/sec}^2$  or  $9.8 \text{ m/sec}^2$ ,  $v_0$  is the initial velocity and  $s_0$  is the initial displacement.

For this problem, a person standing on the edge of a cliff throws a rock directly upward. It is observed that 2 secs later the rock is at its maximum height (in feet) and that 5 secs after that the rock hits the ground at the base of the cliff.

(a) What is the initial velocity of the rock? \_\_\_\_\_ ft/sec.

(b) How high is the cliff?  $t$  \_\_\_\_\_ ft.

(c) What is the velocity of the rock at time  $t$ ? \_\_\_\_\_ ft/sec.

(d) What is the velocity of the rock when it hits the ground?  
\_\_\_\_\_ ft/sec.

14.(1 pt) Problem 20, Section 3.4 page 135.

Let  $s(t) = t^4 - 4t^3 + 8t$  for  $0 \leq t \leq 4$  denote the position of an object moving along a line. Find

(a) The velocity at time  $t$  \_\_\_\_\_

(b) The acceleration at time  $t$  \_\_\_\_\_

(c) Find where the object starts \_\_\_\_\_

and where it stops \_\_\_\_\_

Find the total distance traveled by the object \_\_\_\_\_

(d) Find where the object is accelerating \_\_\_\_\_

Write the answer above in interval notation. **Note:** If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter  $\infty$  as *infinity* and  $-\infty$  as *-infinity*.

15.(1 pt) If a ball is thrown vertically upward from the roof of 64 foot building with a velocity of 80 ft/sec, its height after  $t$  seconds is  $s(t) = 64 + 80t - 16t^2$ . What is the maximum height the ball reaches? \_\_\_\_\_

What is the velocity of the ball when it hits the ground (height 0)? \_\_\_\_\_

16.(1 pt) Problem 18, Section 3.4 page 135.

Let  $s(t) = 3t^2 + 2t - 5$  for  $0 \leq t \leq 1$  denote the position of an object moving along a line. Find

(a) The velocity at time  $t$  \_\_\_\_\_

(b) The acceleration at time  $t$  \_\_\_\_\_

(c) Find where the object starts \_\_\_\_\_

and where it stops \_\_\_\_\_

Find the total distance traveled by the object \_\_\_\_\_

(d) Find where the object is accelerating \_\_\_\_\_

Write the answer above in interval notation. **Note:** If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter  $\infty$  as *infinity* and  $-\infty$  as *-infinity*.

17.(1 pt) Problem 42, Section 3.4 page 136.

Let  $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$  denote the position of an object subject to a constant downward acceleration due to gravity  $g$  (where  $g$  is given approximately by  $32 \text{ ft/sec}^2$  or  $9.8 \text{ m/sec}^2$ ,  $v_0$  is the initial velocity and  $s_0$  is the initial displacement.

For this problem, the projectile is shot upward from the earth with an initial velocity of 320 ft/sec.

- (a) What is the velocity after 5 secs.? \_\_\_\_\_  $ft/sec$ .  
 (b) What is the acceleration after 3 secs.? \_\_\_\_\_  
 $ft/sec^2$ .

**18.**(1 pt) Problem 17, Section 3.4 page 135.

Let  $s(t) = t^2 - 2t + 6$  for  $0 \leq t \leq 2$  denote the position of an object moving along a line. Find

- (a) The velocity at time  $t$  \_\_\_\_\_  
 (b) The acceleration at time  $t$  \_\_\_\_\_

- (c) Find where the object starts \_\_\_\_\_  
 and where it stops \_\_\_\_\_  
 Find the total distance traveled by the object \_\_\_\_\_  
 (d) Find where the object is accelerating \_\_\_\_\_

Write the answer above in interval notation. **Note:** If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter  $\infty$  as *infinity* and  $-\infty$  as *-infinity* .